

1) Define Interrupt Vector Table. Illustrate how to find address of an interrupt service routine corresponding to a given interrupt in 8086.

The interrupt vector (or interrupt pointer) table is the link between an interrupt type code and the procedure that has been designated to service interrupts associated with that code. The interrupt Vector table contains the IP (Instruction Pointer) and CS (Code Segment base address) of all the interrupt types stored sequentially from the address 0000:0000 to 0000:03FFH.

➤ The interrupt type number N is multiplied by 4 and this hexadecimal multiplication gives the offset address of in the code segment with base address 0000H (IVT) at which the IP and CS of the ISR are stored

2. Explain how the INT n instruction finds the starting address of its interrupt service routine in IVT with example.

The first 1Kbyte of memory of 8086 (00000 to 003FF) is set aside as a table for storing the starting addresses of Interrupt Service Procedures (ISP). Since 4-bytes are required for storing starting addresses of ISPs, the table can hold 256 Interrupt procedures. The starting address of an ISP is often called the Interrupt Vector or Interrupt Pointer. Therefore, the table is referred as Interrupt Vector Table. The interrupt type number N is multiplied by 4 and this hexadecimal multiplication gives the offset address of in the code segment with base address 0000H (IVT) at which the IP and CS of the ISR are stored

Eg:- For interrupt 21h, the address in the IVT is $21h * 4 = 0x0024$. Assume the IVT entry at 0x0024 holds 0x1234 (offset) and 0x5678. The complete address of the ISR would be 0x5678:0x1234. The CPU jumps to this address to execute the ISR.

3. Differentiate between maskable and non-maskable interrupts of 8086.

Maskable

The interrupts whose request can be either accepted or rejected by the processor are called maskable interrupts.

When maskable interrupt occur, it can be handled after executing the current instruction.

Maskable interrupts help to handle lower priority tasks.

Maskable interrupts used to interface with peripheral device.

RST6.5, RST7.5, and RST5.5 of 8085 are some common examples of maskable Interrupts.

Non-Maskable

The interrupts whose request has to be definitely accepted (or cannot be rejected) by the processor are called non-maskable interrupts.

When non-maskable interrupts occur, the current instructions and status are stored in stack for the CPU to handle the interrupt.

Non-maskable interrupt help to handle higher priority tasks such as watchdog timer.

Non maskable interrupt used for emergency purpose e.g power failure, smoke detector etc

Trap of 8085 microprocessor is an example for non-maskable interrupt.

4. Explain the stack structure of 8086

The stack is a block of memory that may be used for temporarily storing the contents of the registers inside the CPU. It is a top-down data structure whose elements are accessed using the stack pointer (SP) which gets decremented by two as we store a data word in the stack and gets incremented by two as we retrieve a data word from the stack back to the CPU register.

- The stack segment, like any other segment, may have a memory block of a maximum of 64 Kbytes locations and thus may overlap with any other segments. The stack Segment register (SS) contains the base address of the stack segment in the memory. In 8086 microprocessor-based system, the stack is created by loading a 16-bit base address in the Stack Segment (SS) register and a 16-bit offset address in Stack Pointer (SP). The 20-bit physical address of the stack is computed by multiplying the contents of the SS register by 10H and then adding the contents of SP to this product. Here the content of SP is the offset address of the stack.

5. Define DMA. State the list of operations performed by DMA controller in DMA transfer operation.

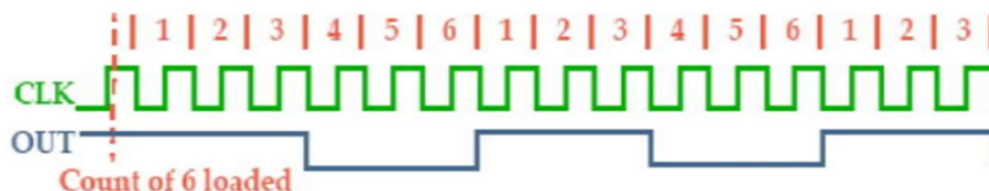
DMA stands for Direct Memory Access. It is designed by Intel to transfer data at the fastest rate. It allows the device to transfer the data directly to/from memory without any interference of the CPU.

Following is the sequence of operations performed by a DMA –

- Initially, when any device has to send data between the device and the memory, the device has to send DMA request (DRQ) to DMA controller.
- The DMA controller sends Hold request (HRQ) to the CPU and waits for the CPU to assert the HLDA.
- Then the microprocessor tri-states all the data bus, address bus, and control bus. The CPU leaves the control over bus and acknowledges the HOLD request through HLDA signal.
- Now the CPU is in HOLD state and the DMA controller has to manage the operations over buses between the CPU, memory, and I/O devices.

6. Discuss how 8254 is used as square wave generator.

Mode 3: Also called square wave generator. When the count N is loaded is even, then for the half of the count the output remains high and rest of the half it remains low. If the count loaded is odd, first clock cycle decrements it by 1 resulting in even count. Then output remains high for first half and low for remaining half. The process is repeated continuously, resulting in generation of square wave. In case of odd count, the output is high for longer duration and low for shorter duration. In general, if loaded count value N is odd it remains high for $(N+1)/2$ pulses and low for $(N-1)/2$ pulses.



7. Justify the statement: “DMA controlled data transfer faster than CPU controlled data transfer”.

Suppose any device which is connected to input-output port wants to transfer data to memory, first of all it will send input-output port address and control signal, input-output read to input-output port, then it will send memory address and memory write signal to memory where data has to be

transferred. In normal input-output technique the processor becomes busy in checking whether any input-output operation is completed or not for next input-output operation, therefore this technique is slow. This problem of slow data transfer between input-output port and memory or between two memory is avoided by implementing Direct Memory Access (DMA) technique. This is faster as the microprocessor/computer is bypassed and the control of address bus and data bus is given to the DMA controller.

8. Write the function of the following control signals in 8255. RD', WR', Ao. A1, RESET, CS'

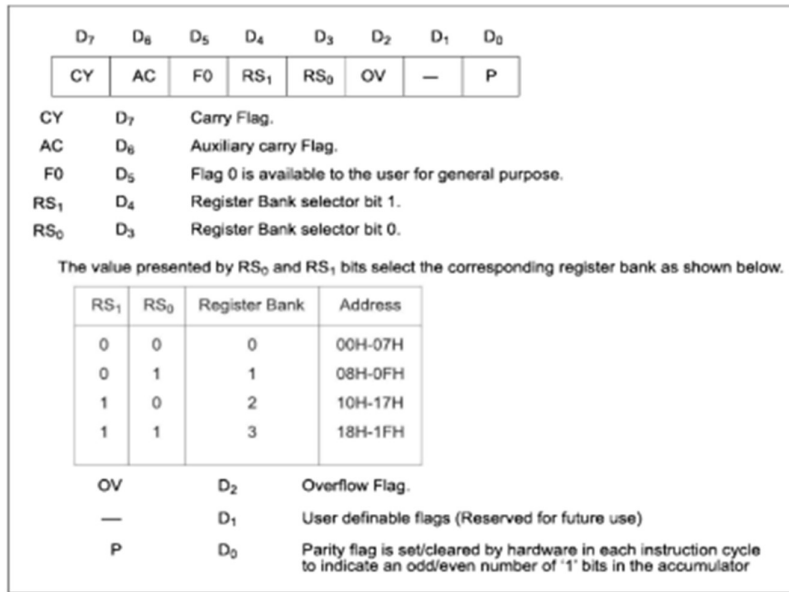
$\overline{RD}$	This input line is driven by the microprocessor and should be low to indicate read operation
$\overline{WR}$	For write operation it should be low
$\overline{CS}$	This is a chip select line. When it goes low it responds to RD/WR signals, otherwise neglected
A ₁ – A ₀	These are address lines driven by the microprocessor. These along with RD/ WR/CS forms the following operations. These address lines are used for addressing any of 4 registers as shown in below table.
RESET	A logic high on this line clears the control word register of 8255. All ports are set as input ports by default after reset.

9. Differentiate between microprocessors and microcontrollers.

MICROPROCESSORS	MICROCONTROLLERS
Microprocessor contains ALU, General purpose registers, stack pointer, program counter, clock timing circuit, interrupt circuit	Microcontroller contains the circuitry of microprocessor, and in addition it has built in ROM, RAM, I/O Devices, Timers/Counters etc.
It has many instructions to move data between memory and CPU	It has few instructions to move data between memory and CPU
Few bit handling instruction	It has many bit handling instructions
Less number of pins are multifunctional	More number of pins are multifunctional
Single memory map for data and code (program)	Separate memory map for data and code (program)
Access time for memory and IO are more	Less access time for built in memory and IO.
Microprocessor based system requires additional hardware	It requires less additional hardwares
More flexible in the design point of view	Less flexible since the additional circuits which is residing inside the microcontroller is fixed for a particular microcontroller
Large number of instructions with flexible addressing modes	Limited number of instructions with few addressing modes

10. Explain program status word (PSW) of 8051 microcontroller.

Program Status Word: The PSW register contains program status information. It is an 8-bit flag register, out of 8-bits 6 bits are used and 2 bits are reserved. Out of 6 bits 4 bits are conditional bits and 2 bits are used for selecting the register bank.



11. with a neat diagram, explain the stack structure of 8086

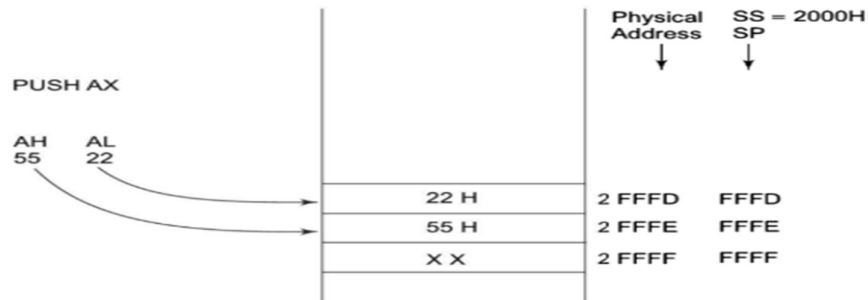
- The stack is a block of memory that may be used for **temporarily storing the contents of the registers** inside the CPU. It is a **top-down data structure whose elements are accessed using the stack pointer (SP)** which gets decremented by two as we store a data word in the stack and gets incremented by two as we retrieve a data word from the stack back to the CPU register.
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SS	⇒	5000 H				
SP	⇒	2050 H				
SS	⇒		0101	0000	0000	0000
10H * SS	⇒		0101	0000	0000	0000
	+					
SP	⇒			0010	0000	0101 0000
Stack-top address			0101	0010	0000	0101 0000
			5	2	0	5 0

- PUSH:** Push to stack

The PUSH instruction decrements the stack pointer (SP) by 2 and copies a word from a specified source to the location in the stack segment to which the stack pointer points.

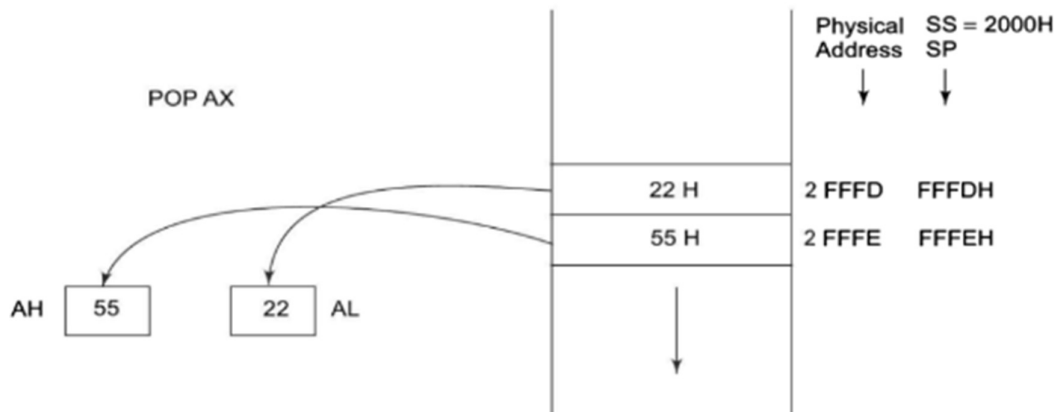
Eg: MOV AX, 5522H



- POP: Pop from stack

The POP instruction copies a word from the stack location pointed to by the stack pointer to a destination specified in the instruction.

Eg;- PUSH AX



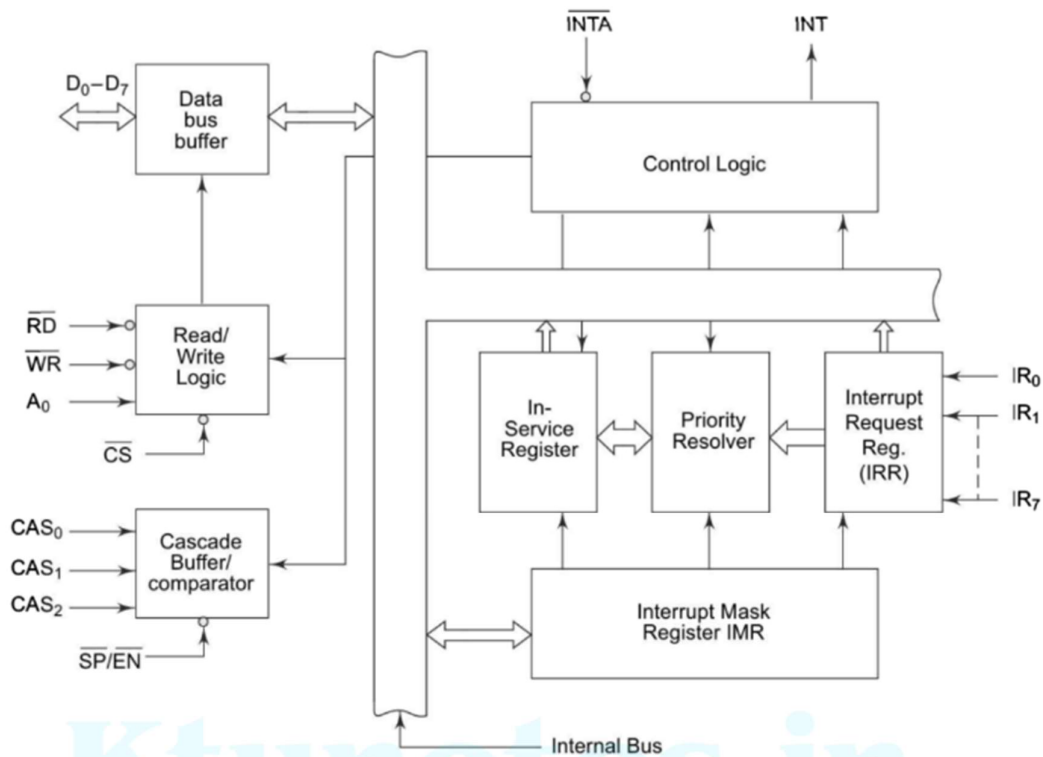
12. List the events performed in 8086 when an interrupt occurs.

The Operation of an Interrupt sequence on the 8086 Microprocessor:

1. External interface sends an interrupt signal, to the Interrupt Request (INTR) pin, or an internal interrupt occurs.
2. The CPU finishes the present instruction (for a hardware interrupt) and sends Interrupt Acknowledge (INTA) to hardware interface.
3. The interrupt type N is sent to the Central Processor Unit (CPU) via the Data bus from the hardware interface.
4. The contents of the flag registers are pushed onto the stack.
5. Both the interrupt (IF) and (TF) flags are cleared. This disables the INTR pin and the trap or single-step feature.

6. The contents of the code segment register (CS) are pushed onto the Stack.
7. The contents of the instruction pointer (IP) are pushed onto the Stack.
8. The interrupt vector contents are fetched, from $(4 \times N)$ and then placed into the IP and from $(4 \times N + 2)$ into the CS so that the next instruction executes at the interrupt service procedure addressed by the interrupt vector.
9. While returning from the interrupt-service routine by the Interrupt Return (IRET) instruction, the IP, CS and Flag registers are popped from the Stack and return to their state prior to the interrupt.

13. With a neat diagram explain the architecture of 8259, Programmable interrupt controller.



Data Bus Buffer This tristate bidirectional buffer interfaces internal 8259A bus to the microprocessor system data bus. Control words, status and vector information pass through buffer during read or write operations.

Read /Write Control Logic This circuit accepts and decodes commands from the CPU. This also allows the status of the 8259A to be transferred onto the data bus.

Interrupt Request Register (IRR): The interrupts at IRQ input lines are handled by Interrupt Request Register internally. IRR stores all the interrupt requests in it in order to serve them one by one on a priority basis.

Priority Resolver This unit determines the priorities of the interrupt requests appearing simultaneously. The highest priority is selected and stored in the corresponding bit of ISR during the INTA pulse. The IRO has the highest priority while the IR7 has the lowest one.

In-Service Register (ISR) This stores all the interrupt requests those are being served, i.e. ISR keeps a track of the requests being served.

Interrupt Mask Register (IMR) This register stores the bits required to mask the interrupts.

Control Logic This block manages the interrupt and interrupt acknowledge signals to be sent to the CPU for serving one of the eight interrupt requests. This also accepts interrupt acknowledge (INTA) signal from the CPU (8086) that causes the 8259A to release vector address onto the data bus.

Cascade Buffer/Comparator This block stores and compares the ID's of all the 8259As used in the system in a cascade configuration. The three I/O pins CASO - CAS2 act as output when the 8259A is used as a master. The same pins act as input when 8259 is in slave mode.

14. Design an interface between 8086 and two 4K x 8 EPROMs and two 4K x 8 RAM chips. Select suitable address maps.

Memory Map Table

Address	A19	A18	A17	A16	A15	A14	A13	A12	A11	A10	A09	A08	A07	A06	A05	A04	A03	A02	A01	A00
FFFFH	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
EPROM								8K X 8												
FE000H	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0
FDFFFH	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1
RAM								8K X 8												
FC000H	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Total 8K bytes of EPROM need 13 address lines A0-A12 (since $2^{13} = 8K$).

Address lines A13 - A19 are used for decoding to generate the chip select.

The ___ signal goes low when a transfer is at odd address or higher byte of data is to be accessed.

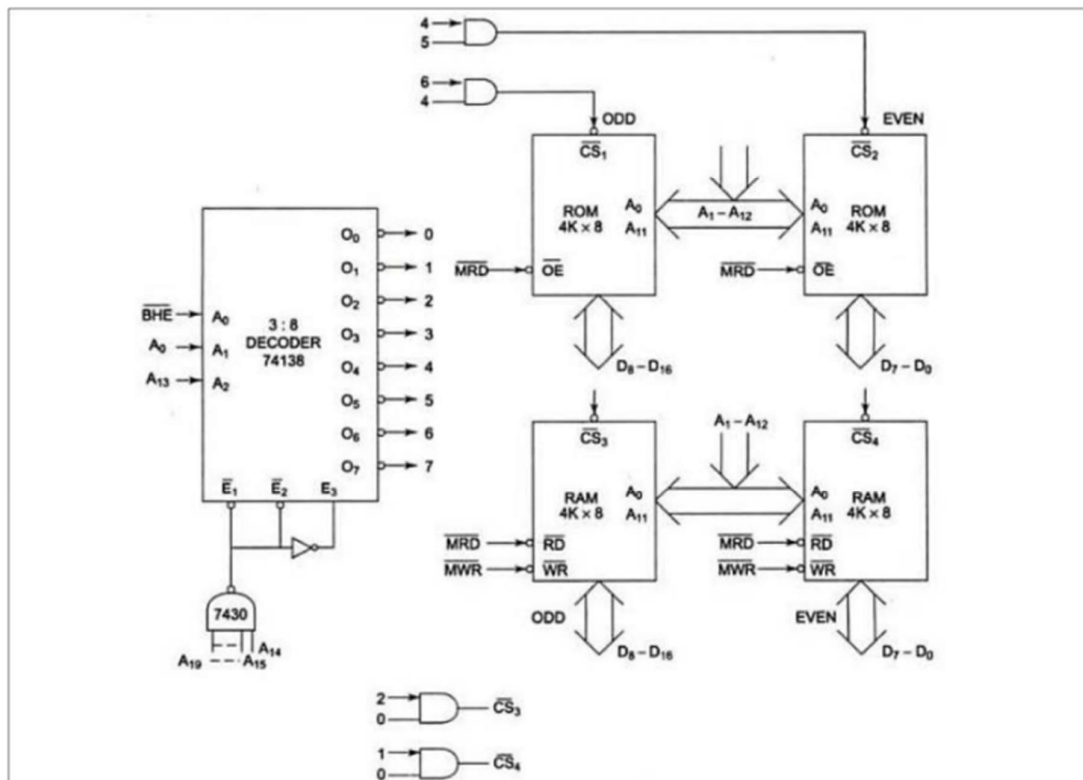
Let us assume that the latched address, ___ and demultiplexed data lines are readily available for interfacing.

The memory system in this problem contains in total four 4K x 8 memory chips.

The two 4K x 8 chips of RAM and ROM are arranged in parallel to obtain 16-bit data bus width. If A0 is 0, i.e., the address is even and is in RAM, then the lower RAM chip is selected indicating 8-bit transfer at an even address. If A0 is 1, i.e., the address is odd and is in RAM, the ___ goes low, the upper RAM chip is selected, further indicating that the 8-bit transfer is at an odd address. If the selected addresses are in ROM, the respective ROM chips are selected. If at a time A0 and ___ both are 0, both the RAM or ROM chips are selected, i.e., the data transfer is of 16 bits. The selection of chips here takes place as shown in table below.

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Memory Chip Selection Table:

Decoder I/P --> Address/ <u>BHE</u> -->	A2 A13	A1 A0	A0 <u>BHE</u>	Selection/ Comment
Word transfer on D0 - D15	0	0	0	Even and odd address in RAM
Byte transfer on D7 - D0	0	0	1	Only even address in RAM
Byte transfer on D8 - D15	0	1	0	Only odd address in RAM
Word transfer on D0 - D15	1	0	0	Even and odd address in RAM
Byte transfer on D7 - D0	1	0	1	Only even address in RAM
Byte transfer on D8 - D15	1	1	0	Only odd address in ROM

15. Explain the operating modes of 8255.

1. INPUT/OUTPUT MODE

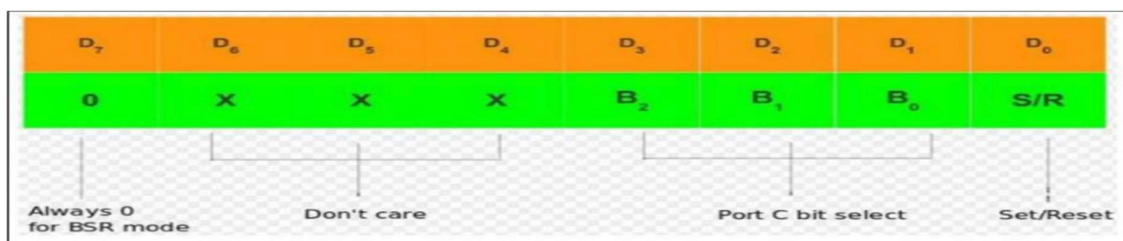
There are three basic modes of operation than can be selected by the system software:

- **Mode 0 -Basic Input/Output**
 - **Mode 1 -Strobed Input/Output**
 - **Mode 2 -Bi-directional Bus**
-
- **Mode 0** – In this mode, Port A and B is used as two 8-bit ports and Port C as two 4-bit ports. Each port can be programmed in either input mode or output mode where outputs are latched and inputs are not latched. Ports do not have interrupt capability.
 - **Mode 1** - In mode 1 each group can be used as 8-bit input or output data bus and the remaining 4-bits are used as handshaking and interrupt control signals. Port A is used with port c upper three bits and port B is used with port c lower 3 bits. The remaining 2 bits of port C can be used as control signals.
 - **Mode 2** – In this mode, Port A can be configured as the bidirectional port and Port B either in Mode 0 or Mode 1. Port A uses five signals from Port C as handshake signals for data transfer. The remaining three signals from Port C can be used either as simple I/O or as handshake for port B.

2. BIT SET/RESET (BSR) MODE

In this mode only port b can be used (as an output port). Each line of port C (PC0 - PC7) can be set/reset by suitably loading the command word register.no effect occurs in input-output mode. The individual bits of port c can be set or reset by sending the signal OUT instruction to the control register.

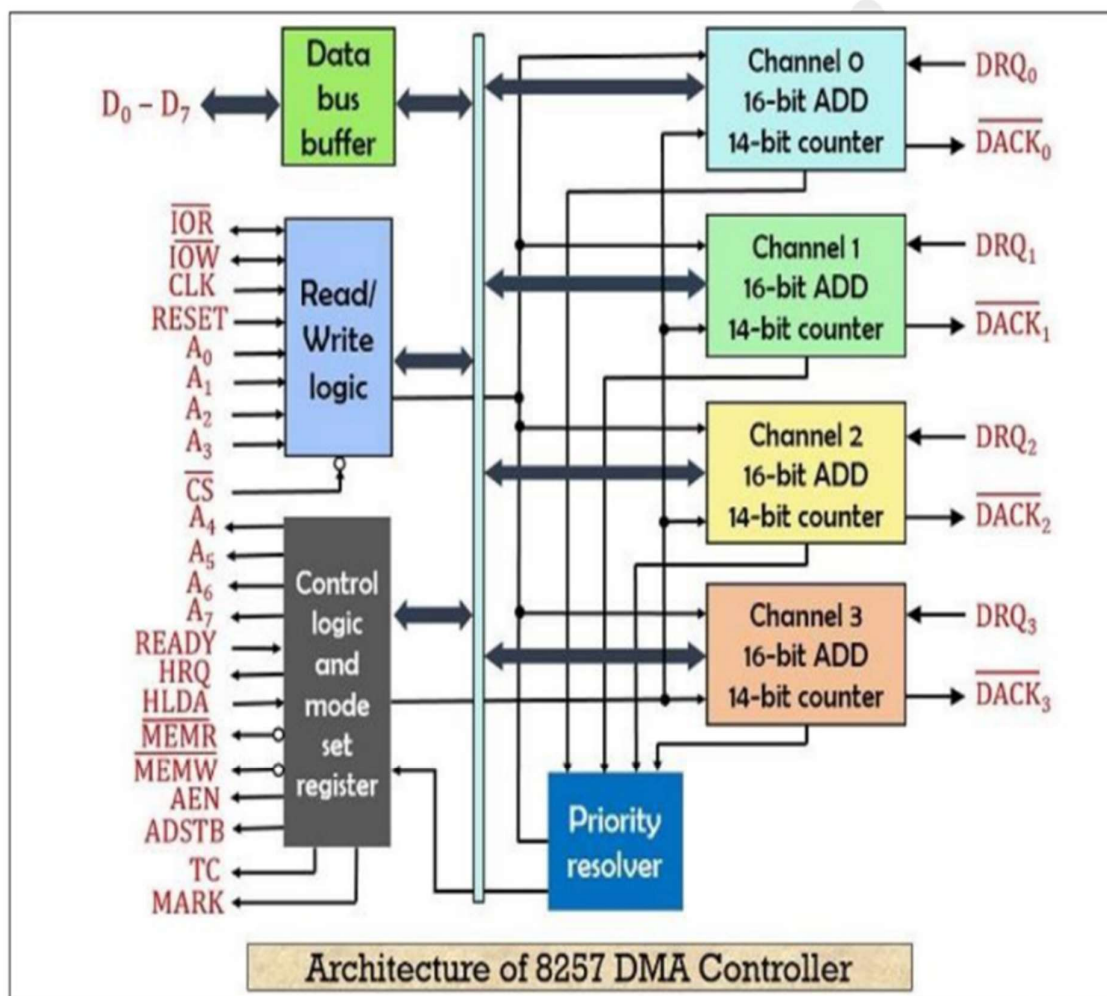
- The figure shows the control word format in BSR mode. This mode is selected by making D7='0'.
- D0 is used for bit set/reset. When D0= '1', the port C bit selected (selection of a port C bit is shown in the next point) is SET, when D0 = '0', the port C bit is RESET.
- D1, D2, D3 are used to select a particular port C bit whose value may be altered using D0 bit as mentioned above.



16. With a neat diagram, explain the features and architecture of 8257, DMA controller

Here is a list of some of the prominent features of 8257 –

- It has four channels which can be used over four I/O devices.
- Each channel has 16-bit address and 14-bit counter.
- Each channel can transfer data up to 64kb.
- Each channel can be programmed independently.
- Each channel can perform read transfer, write transfer and verify transfer operations.
- It generates MARK signal to the peripheral device that 128 bytes have been transferred.
- It requires a single-phase clock.
- Its frequency ranges from 250Hz to 3MHz.
- It operates in 2 modes, i.e., Master mode and Slave mode.



Data Bus Buffer:

8-bit Tristate, bidirectional buffer interfaces the internal bus of 8257 with the external system bus under the control of various control signals.

Read/Write Logic:

In the slave mode, the read/write logic accepts the I/O Read or I/O Write signals, decodes the A0-A3 lines and either writes the contents of the data bus to the addressed internal register or reads the selected register depending upon whether IOW or IOR signal is activated. In master mode, the read/write logic generates the IOR and IOW signals to control the data flow to or from the selected peripheral.

Control Logic:

The control logic controls the sequences of operations and generates the required control signals like AEN, ADSTB, MEMR, MEMW, TC and MARK along with the address lines A4-A7, in master mode.

Priority Resolver:

The priority resolver resolves the priority of the four DMA channels depending upon whether normal priority or rotating priority is programmed.

Register Organisation of 8257:

The 8257 performs DMA operation over four independent DMA channels with the following Registers.

1. DMA Address Register

Each DMA channel has one DMA address register. The function of this register is to store the address of the starting memory location, which will be accessed by the DMA channel. The device that wants to transfer data over a DMA channel, will access the block of the memory with the starting address stored in the DMA Address Register.

2. Terminal Count Registers

Each of the four DMA channels of 8257 has one terminal count register (TC). This 16-bit register is used for ascertaining that the data transfer through a DMA channel ceases or stops after the required number of DMA cycles.

After each DMA cycle, the terminal count register content will be decremented by one and finally it becomes zero after the required number of DMA cycles are over. The bits 14 and 15 of this register indicate the type of the DMA operation (transfer).

3. Mode Set Register

The mode set register is used for programming the 8257 as per the requirements of the system. The function of the mode set register is to enable the DMA channels individually and also to set the various modes of operation. The bits Do-D3 enable one of the four DMA channels of 8257. If the TC STOP bit is set, the selected channel is disabled after the terminal count condition is reached, and it further prevents any DMA cycle on the channel. If the TC STOP bit is programmed to be zero, the channel is not disabled, even after the count reaches zero and further request are allowed on the same channel. The auto load bit, if set, enables channel 2 for the repeat block chaining operations, without immediate software intervention between the two successive blocks. The extended write bit, if set to '1', extends the duration of MEMW and IOW signals by activating them earlier, which is useful in interfacing the peripherals with different access times.

4. Status register

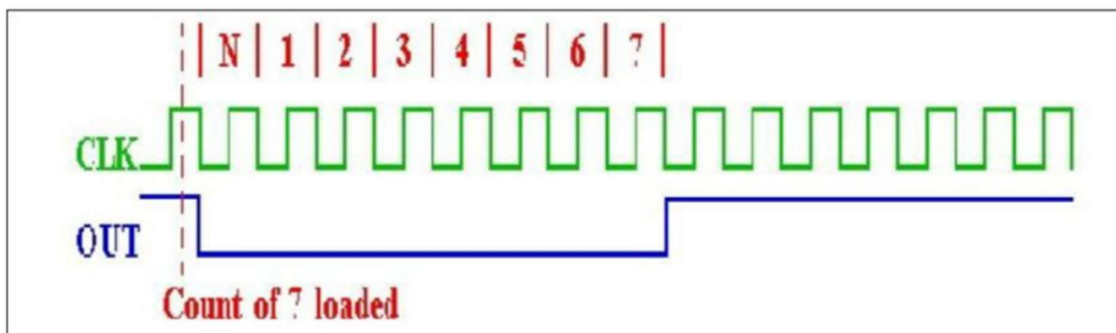
The lower order 4-bits of this register contain the terminal count status for the four individual channels. If any of these bits is set, it indicates that the specific channel has reached the terminal count condition. The update flag is not affected by the read operation. This flag can only be cleared by resetting 8257. The update flag is set every time, the channel 2 registers are loaded with contents of the channel 3 registers. It is cleared by the completion of the first DMA cycle of the new block. This register can only read.

17. Explain the operating modes of 8254, programmable interval timer.

8253/54 can be operated in 6 different modes. In this chapter, we will discuss these operational modes.

i. Mode 0 – Interrupt on Terminal Count

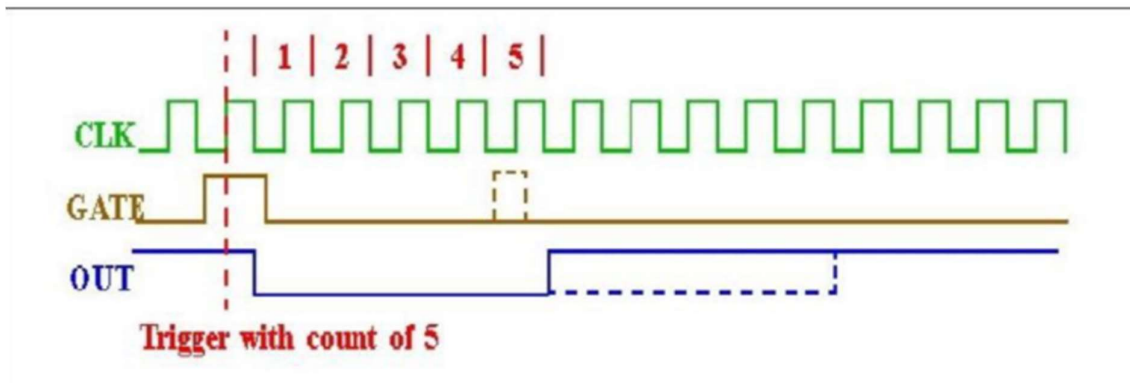
It is used to generate an interrupt to the microprocessor after a certain interval. Initially the output is low after the mode is set. The output remains LOW after the count value is loaded into the counter.



ii. Mode 1 – Programmable One Shot

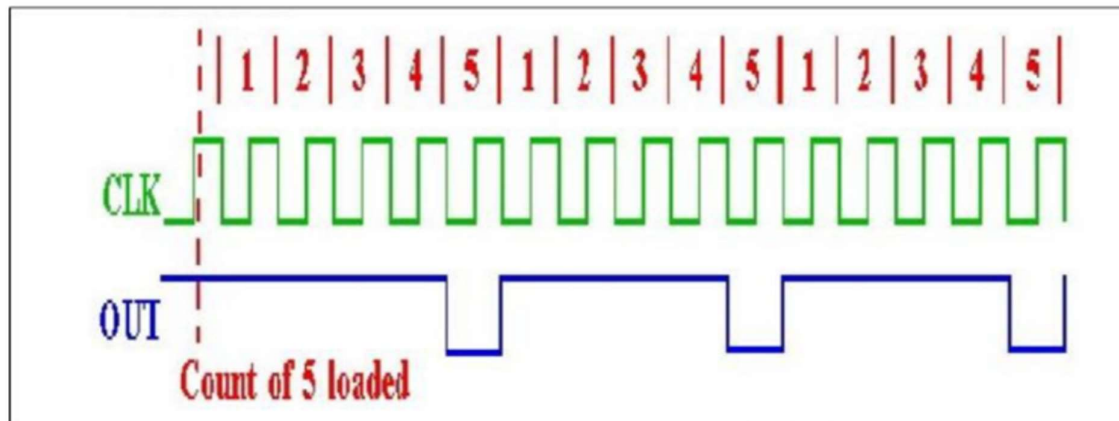
It can be used as a mono stable multi-vibrator. The gate input is used as a trigger input

in this mode. The output remains high until the count is loaded and a trigger is applied.



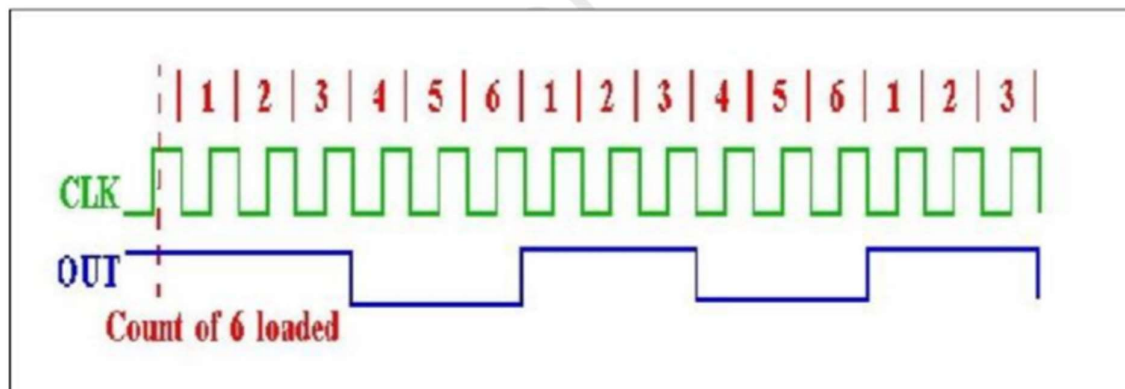
iii. Mode 2 – Rate Generator

The output is normally high after initialization. Whenever the count becomes zero, another low pulse is generated at the output and the counter will be reloaded.



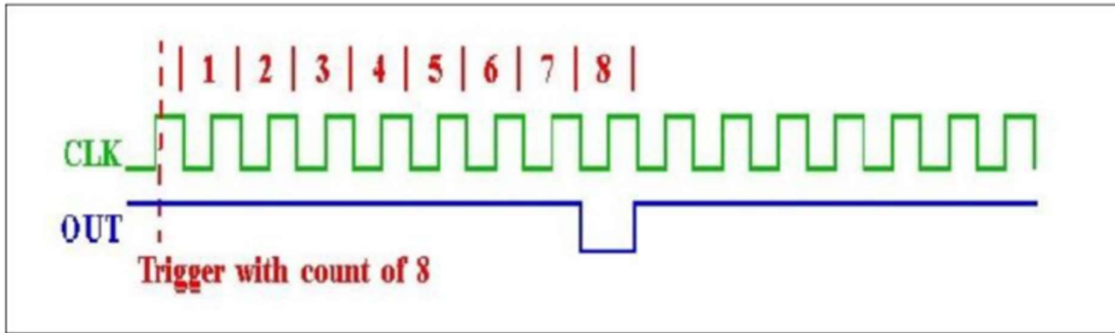
iv. Mode 3 – Square Wave Generator

This mode is similar to Mode 2 except the output remains low for half of the timer period and high for the other half of the period.



v. Mode 4 – Software Triggered Mode

In this mode, the output will remain high until the timer has counted to zero, at which point the output will pulse low and then go high again. The count is latched when the GATE signal goes LOW. On the terminal count, the output goes low for one clock cycle then goes HIGH. This low pulse can be used as a strobe.



vi. Mode 5 – Hardware Triggered Mode

This mode generates a strobe in response to an externally generated signal. This mode is similar to mode 4 except that the counting is initiated by a signal at the gate input, which means it is hardware triggered instead of software triggered. After it is initialized, the output goes high. When the terminal count is reached, the output goes low for one clock cycle.

